

Option Trading

Session Three: Options Pricing Models and Greeks

About me

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Let's start learning!

Session Three Overview

- Options basics and terminology.
- Model independent features of options: arbitrage relationships between various options, and options and underlying.
- Option pricing variables and parameters.
- Properties of BSM solution.
- Options Greeks.

Options

Basics and Terminology

Question-1

**Have you gone through the
options primer?**

Options

- An option gives the holder the right but not the obligation to do something.
- A call confers the right to buy the underlying at a certain price, the strike price.
- A put confers the right to sell the underlying at a certain price, the strike price.

Options

- Example: $S = \$100$, a 90-strike call happens to be worth \$14.
- \$10 dollars of this is intrinsic value.
- The remaining \$4 is extrinsic or time value.
- The intrinsic value is the amount by which the strike price of an option is profitable as compared to the stock's price in the market.
- Options also have “time value” (or “extrinsic value”) which reflects the fact the something might happen in the future to make the option intrinsically valuable.

Options

- In-the-money (ITM) means an option would have value if it expired right now. So, a call with a strike lower than the underlying price is ITM.
- We call this amount as 'Intrinsic Value'
- An out-of-the-money (OTM) option has no intrinsic value.
- An at-the-money (ATM) option has a strike equal to the underlying price (or sometimes the forward price so check in specific cases).

Options

- European options can only be exercised at a single time, the expiration date.
- American options can be exercised at a time prior to the expiration date.
- Almost all exchange listed options are American.
- The concept of European options was invented by Black and Scholes, because they had an analytic solution in this case, but these fixed-term options weren't like the ones that actually traded in America.

Options

- There are also Asians, which settle to an average.
- Russians, which are eternal lookback options.
- Bermudans, which can be exercised at a set of specific dates.
- Parisians, which are barriers that kick in when underlying has spent a certain time beyond the barrier.
- Hawaiians, an Asian option with an American exercise feature.

Exotic Options

- Many other exotic options “exist”.
- But most rarely trade or have never even been traded at all.
- Mainly an example of the worst kind of financial mathematics.
- Even when they exist, their “trading” generally consists of one-time selling to a customer (at an enormous markup).

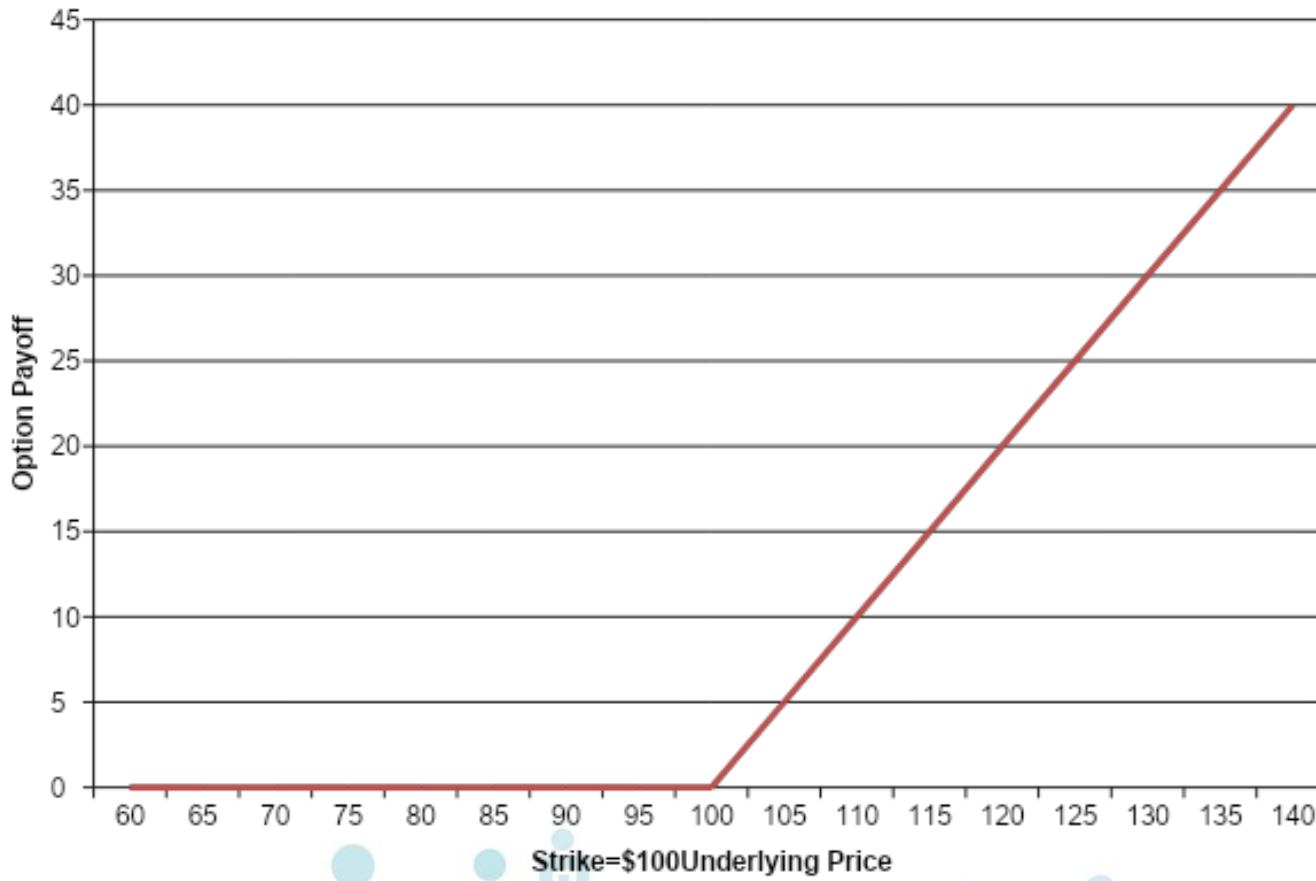
Options and Trading

- General rule: simplest products trade in the most complicated ways.
- Futures: market makers use many “tricks” such as book stacking, spoofing, flipping and stop hunting.
- Vanilla options: capturing edge is easy and traders are differentiated by risk management.
- Exotic Options: edge comes from structuring, legal and sales.

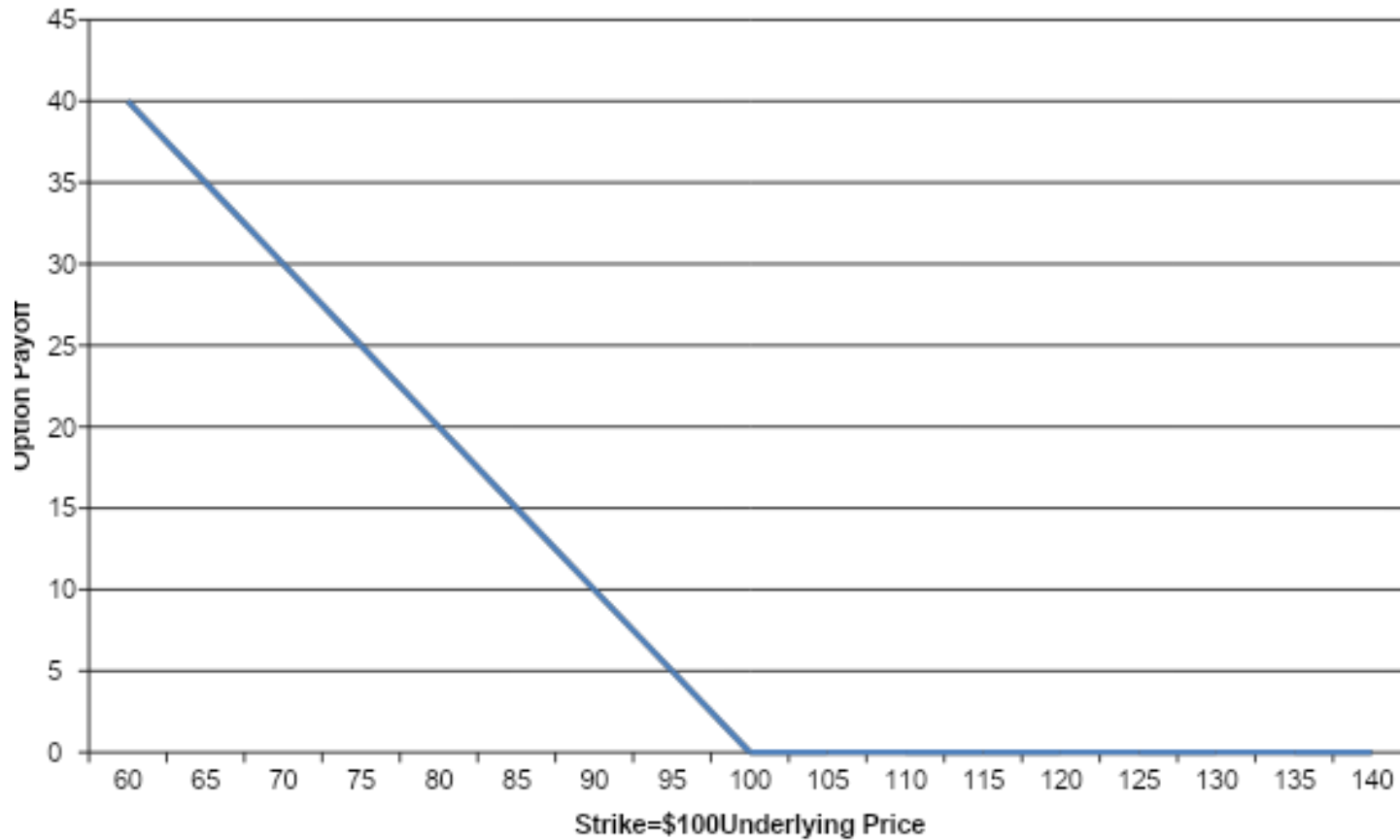
Option Payoff Functions

- The payoff is the value of the option at expiration.
- If the underlying price is S and the strike is X :
- Call payoff is $\text{MAX}(S-X, 0)$
 - So, the holder makes money if the stock goes above the strike.
- Put payoff is $\text{MAX}(X-S, 0)$
 - So, the holder makes money if the stock goes below the strike.

Payoff Diagram: A Call



Payoff Diagram: A Put



Combinations

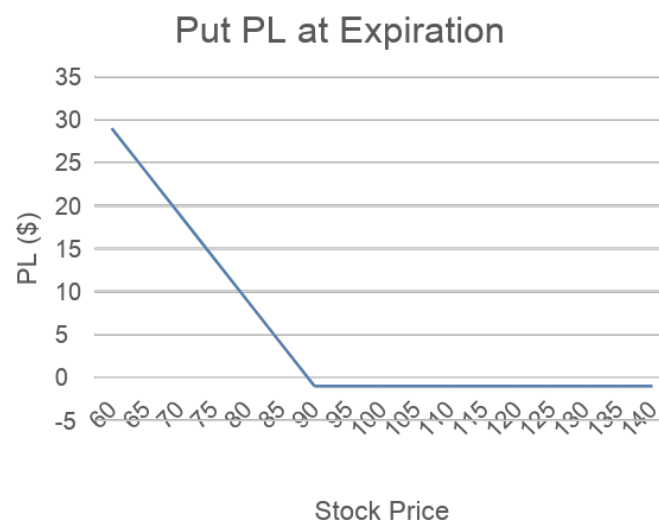
- The piecewise linearity of options means it is easy to construct combinations of puts and calls to express any view.
- Aside: Be very careful about overusing this feature.
- Being right on a single view is hard enough.

Options

Trade Examples

Trade Example: Hedging

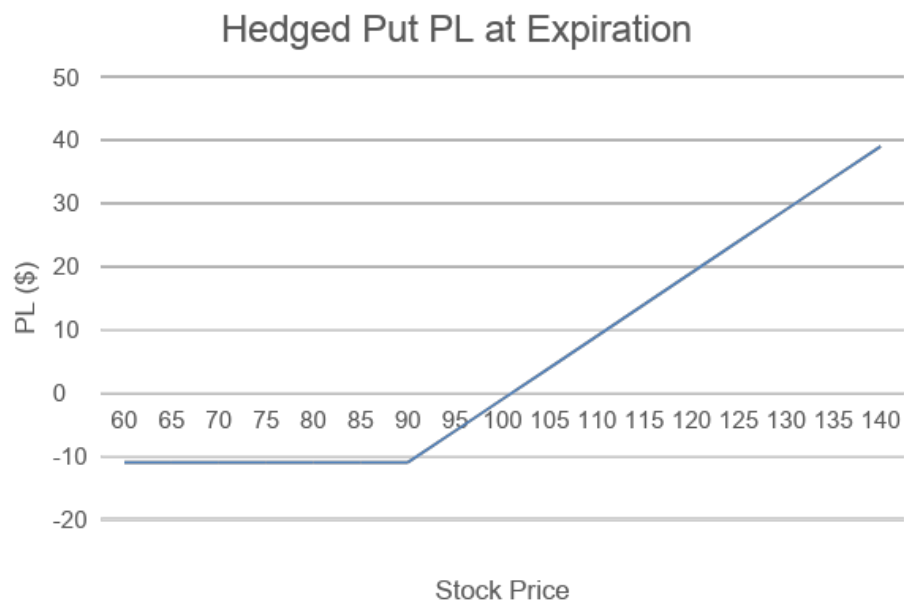
- Using options to limit downside losses.
- Example: long a share at \$90. Pay \$10 for the 90 put.



- Losses are guaranteed to be capped, unlike a sell order.

Trade Example: Hedging

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Directional Option Trading

Pros:

- Limited downside (you can lose only the premium paid for option).
- Unlimited upside (literally, in the case of calls, and practically in the case of puts).

Directional Option Trading

Cons:

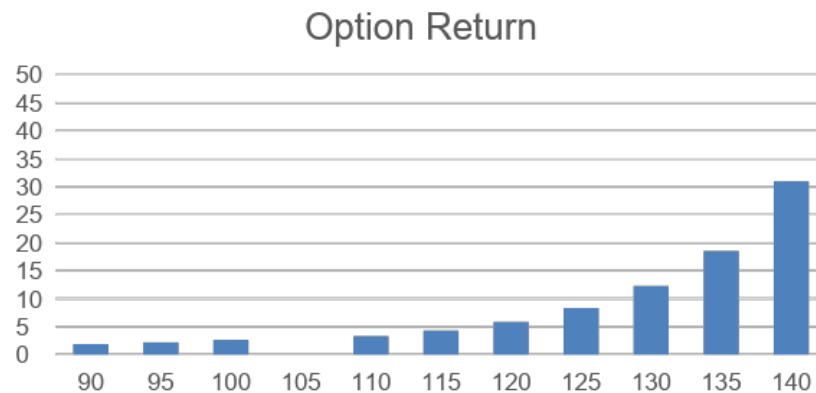
- Trade has limited time to work out.
- Can choose wrong strike.
- High transaction costs.
- Usually pay more than fair value.
- (Can really only manage the first two.)

Strike Choice

- Return is highest when you pick the strike that only just expires in the money.
- But:
 - We don't know what that will be.
 - If we are wrong the option expires worthless.

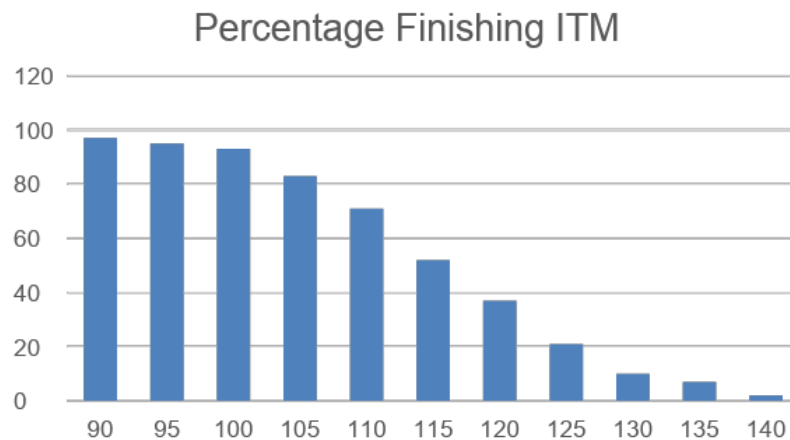
Strike Choice

- Example: one-year options on a \$100 stock.
- Average return is 15% and volatility is 10%.



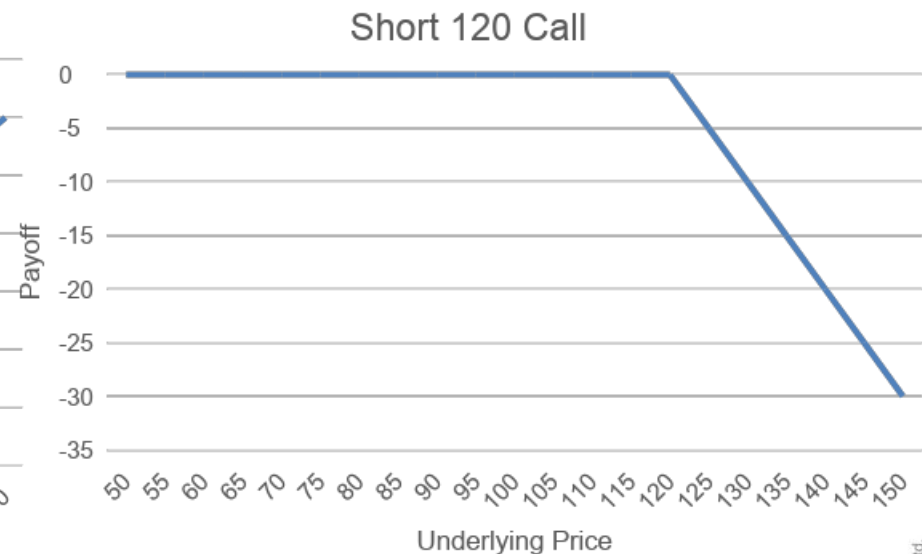
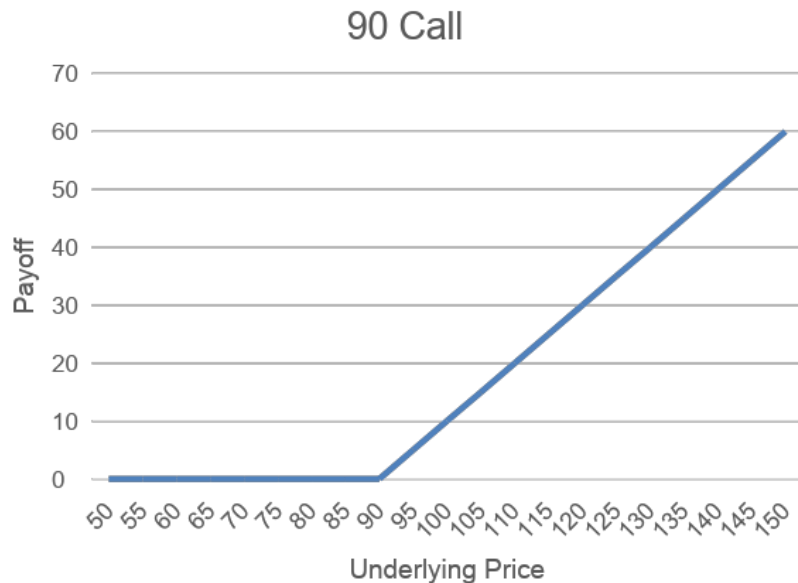
Strike Choice

- But higher strikes are more often worthless.



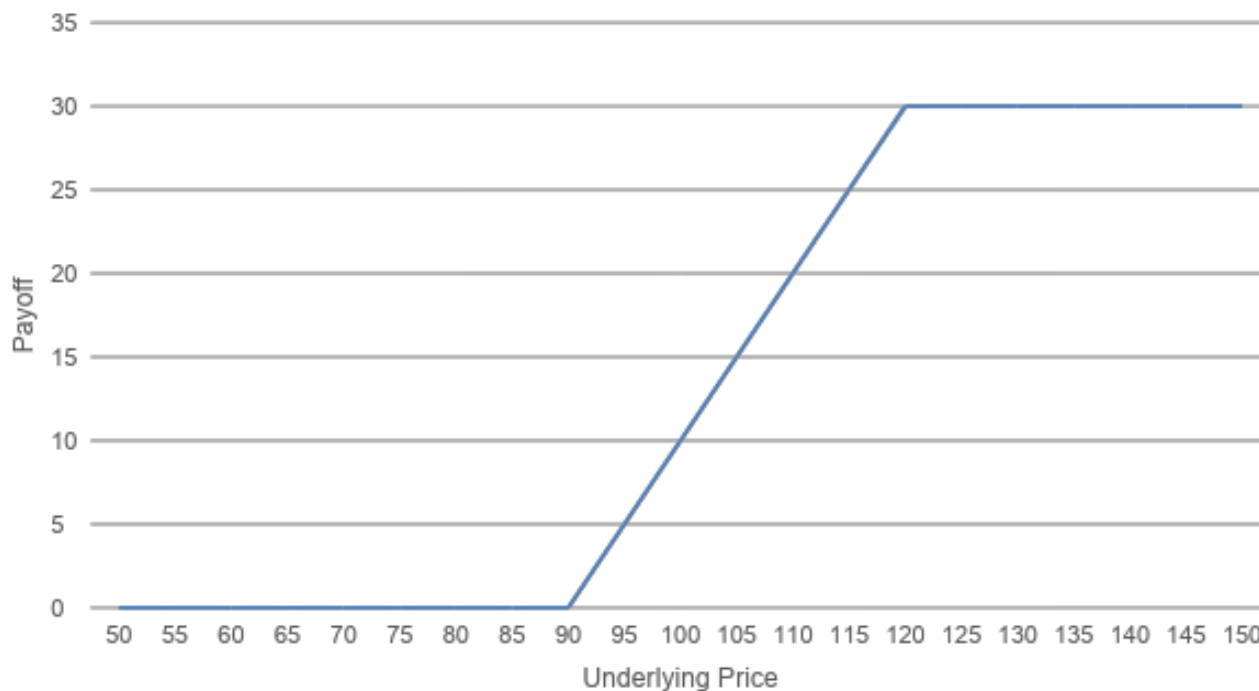
Example: Call Spread

- A long call spread is made by buying a call and selling a call with a higher strike and the same expiration.



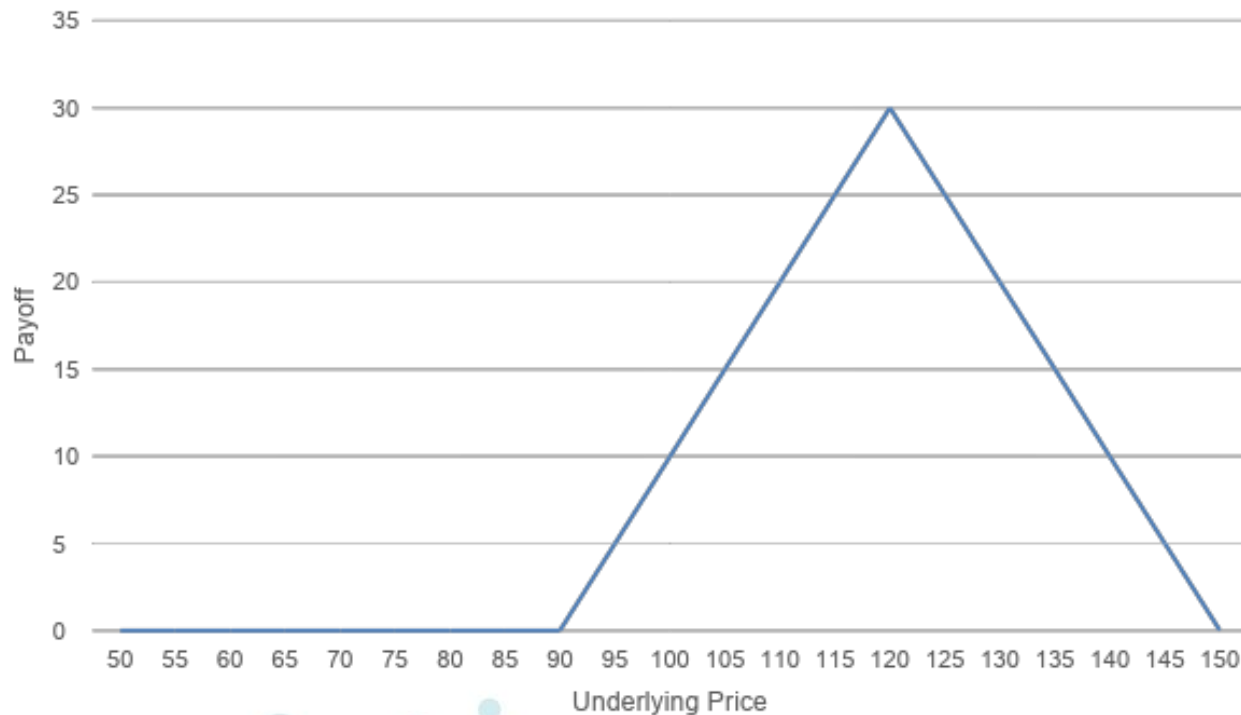
Example: Call Spread

- A long call spread is made by buying a call and selling a call with a higher strike and the same expiration.



Example: 1 by 2 Call Spread

- A long 1 by 2 call spread is made by buying a call and selling two calls with a higher strike and the same expiration.



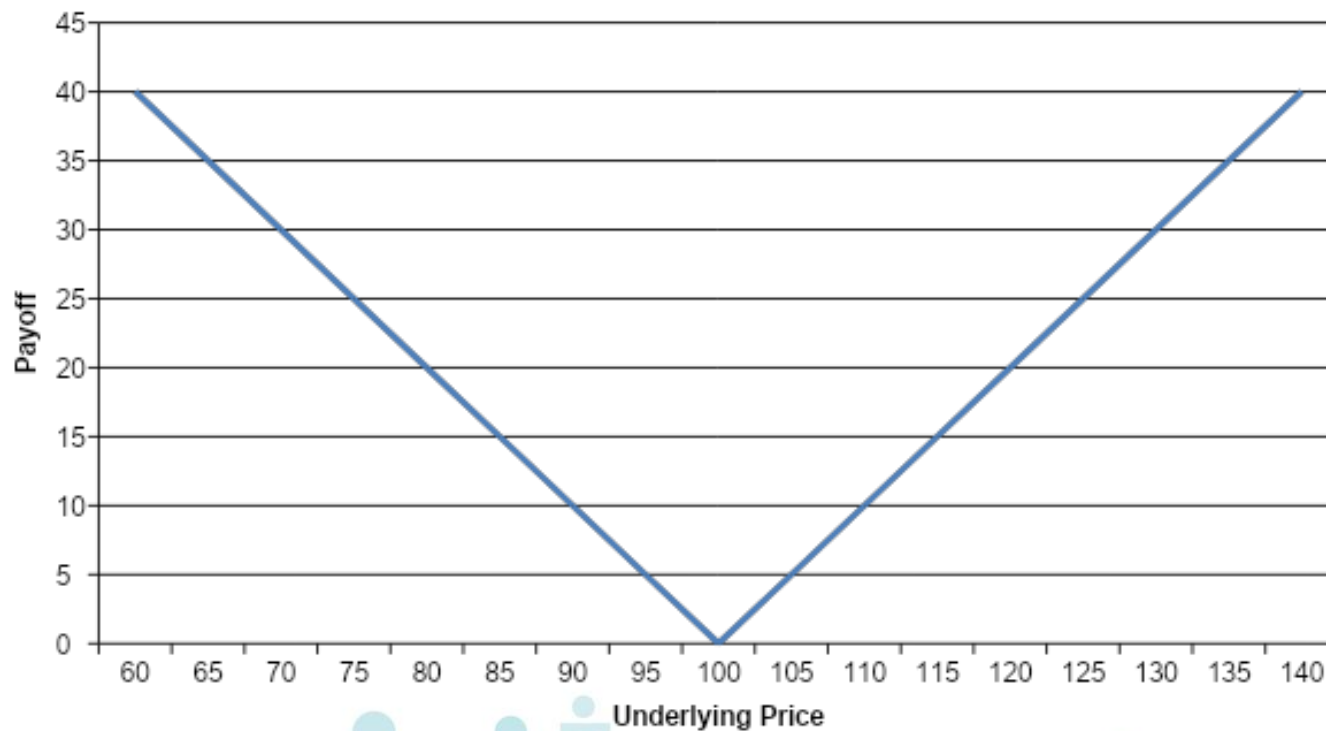
Example: Risk-Reversal

- A risk-reversal is made by buying a put and selling a call with a higher strike and the same expiration.



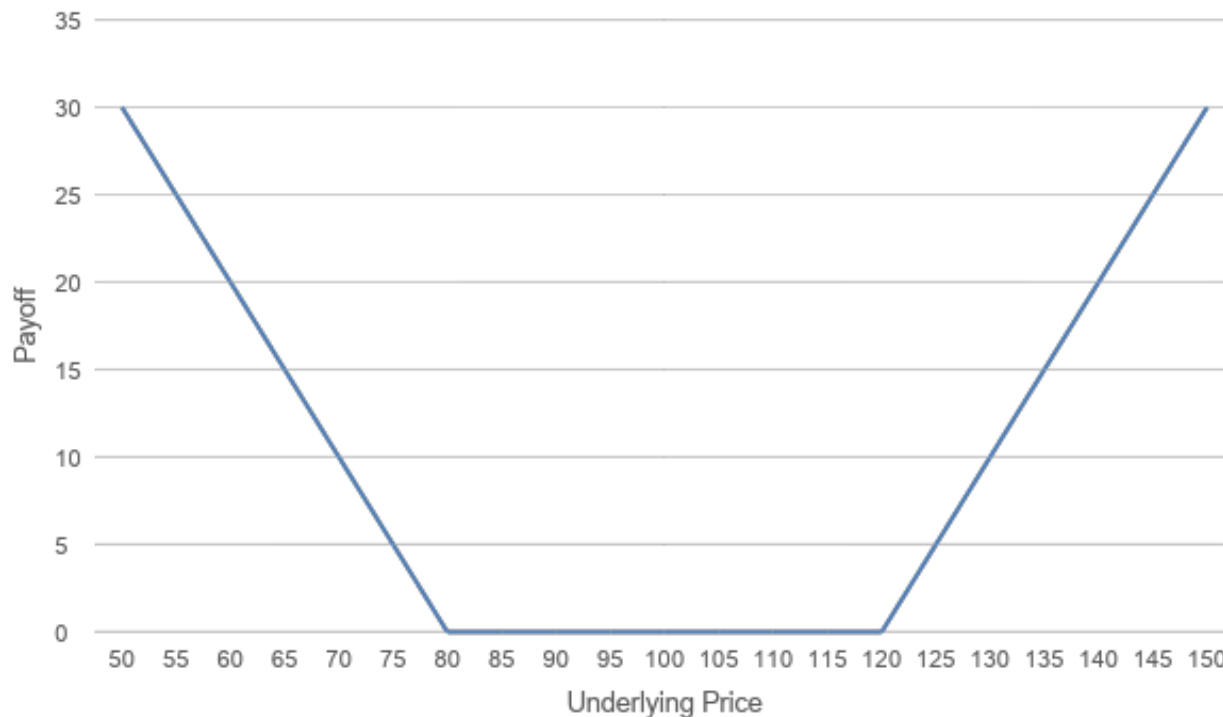
Example: Straddle

- A long straddle is made by buying a put and a call with the same strike and expiration.



Example: Strangle

- A long strangle is made by buying a put and a call with different strikes and same expiration.



Options

Model independent features of options

Model Independent Relationships

American > European

- American options are worth more than Europeans.
- If you don't choose to exercise an American option IS a European option.
- So, it is European plus some extra value.

Longer Dated American Options > Short Dated

- If you feel like it, you can turn the longer dated option into the short one by exercising.
- So, it IS the short-dated option plus some more optionality.
- Usually but not always true for European options (in high-rate environments getting cash sooner is better so a short-dated put can be worth more).

Call < Underlying

- If you exercise, you get the underlying.
- So, if $C > S$ just sell the call and buy the underlying.
- At expiration our profit is C-expiration value of call + profit on stock.
- Above the strike this is $C - (S - X) + (S - S_0) = C + (X - S_0) > X$
- Below the strike this is $C - 0 + (S - S_0) > 0$

Call<Underlying

- Example: 100 strike call on \$100 stock is trading for \$102.
- Sell call and buy stock.
- Assume we expire at \$110. Call exercised.
- Lose \$10 due to exercise, keep \$102 call premium, gain \$10 from our stock purchase.
- Make \$102.
- Assume we expire at \$90.
- We lose \$10 on our stock purchase but keep \$102 from call sale.

Put < Strike

- If you exercise, you get the strike minus the current price.
- So, the highest value the option can ever have is the strike (if $S=0$).
- Example: 100 strike put on \$100 stock is trading for \$102. Sell the put.
- Expire at \$0. Lose \$100 on put exercise but keep \$102 put premium.
- Expire above the strike, keep \$102 premium.

Lower Strike C > Higher Strike C

- There will be times when a lower strike call can be exercised, and the higher strike call cannot.
- Example: 100 strike call is trading for \$2 and \$110 call is trading for \$5.
- Sell 110 call, buy 100 call and collect \$3.
- Expire at \$90, both calls are worthless, and we keep \$3.
- Expire at \$105. Make \$5 on the 100 call and keep \$3 premium.
- Expire at \$120. Make \$20 on 100 call, lose \$10 on 110 call and keep \$3 premium.
- Opposite is true for puts.

Options

Put/Call Parity

Put/Call Parity

- Puts and calls are “the same thing”
- Specifically

$$C - P = S - X e^{-rT}$$

- Proof: Form a portfolio of long put, short call and long one share.

Instrument	Initial Value	Value at Expiration	
		If $S < X$	If $S > X$
Long P(X)	P(X)	(X-S)	0
Short C(X)	-C(X)	0	-(S-X)
Long Stock	S	S	S
Total Portfolio	$S + P(X) - C(X)$	X	X

Put Call Parity Implications

- For call and put to have equal value the strike price must be the forward price.

$$C - P = S - X e^{-rT}$$

- Assume $C = P$

$$0 = S - X e^{-rT}$$

$$X = S e^{rT}$$

Put Call Parity Implications

- The relationship does not hold for American options, because we can't be sure of the duration of a short option position.

$$C - P = S - X e^{-rT}$$

Put/Call Parity - American Options

- For European options

$$C - P = S - X e^{-rT}$$

- Example: $S=0$, $X=100$, $T=1$, $r=0.1$
- $C=0$, $P=90.48$ so $C-P=-90.48$ (from the BSM model)
- $S - X e^{-rT} = -90.48$
- But for Americans, $C=0$ and $P=100$, or else we just exercise, hence P/C parity equation fails.

Put Call Parity Implications

- If we can make a bullish option position into a bearish option position by adding stock, then we should also be able to directionally neutralize an option.
- So, P/C parity is an important precursor to volatility trading.

Options

Pricing Concepts

Option Pricing: Inputs

- Underlying price.
- Strike: distance between strike and price define the option.
- Rates : rates are the discount factor for cash flows.
- Dividends: income from the stock which affects its future value.
- Time.
- Volatility.

Option Pricing: Inputs

- Why not return?
- Surely if $\mu > 0$ calls will be worth more and puts less?
- Refer back to put/call parity.

$$C - P = S - X e^{-rT}$$

- If you believe a positive drift leads C to increase, it also means P must increase.
- So positive drift leads both puts and calls to increase?
- Contradiction.

BSM - Option Value

Closed-form solution of the BSM equation

Theoretical
call option price

Strike Price

Stock Price

$$C = S N(d_1) - X \exp(-rT) N(d_2)$$

$N()$ is cumulative normal distribution.

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

Option Value

- $N(d_2)$ is the amount of cash we are exposed to. i.e. what we need to deliver.
- Alternatively, $N(d_2)$ is the probability we need to pay out on the option.
- $N(d_2)$ is the probability of finishing in the money.
- No!
- Remember all of this takes place in the *risk neutral* world.

Option Value

- This interpretation is only true in a risk-neutral world; a world that doesn't exist.
- In the real world, the probabilities are mainly driven by drift, something we have no interest in (as a pricing variable) or significant ability to predict.
- Never use options to make probabilistic statements about the real prices.
- This has been tested. Options do not finish ITM at the rate predicted by $N(d_2)$.

Option Value

A put can be prices by using the P/C parity

$$P = X \exp(-rT) N(-d_2) - S N(-d_1)$$

BSM - Assumptions

- Continuous Trading
- Constant Volatility
- Log-Normal Distribution
- Risk-Free Interest Rate
- No Dividends
- No Transaction Costs
- European-Style Options
- Efficient Markets

BSM - Practical Applications

- Risk Management
- Delta Hedging
- Volatility Trading
- Arbitrage Opportunities
- Option Strategies

Options

Greeks

“Greeks”

- “Greeks” is the collective term for partial derivatives (in the mathematical sense) of the option price.
- Can be calculated either analytically or numerically.
- Volatility traders generally quantify their positions i.t.o. Greek exposure.
- Because the BSM equation is linear, greeks can be aggregated across a position by adding the greeks of each individual option.

Options

Greeks: Delta

Delta

- Delta is the partial derivative of the option price w.r.t. underlying.

$$\Delta_{call} = N(d_1)$$

$$\Delta_{put} = -N(-d_1)$$

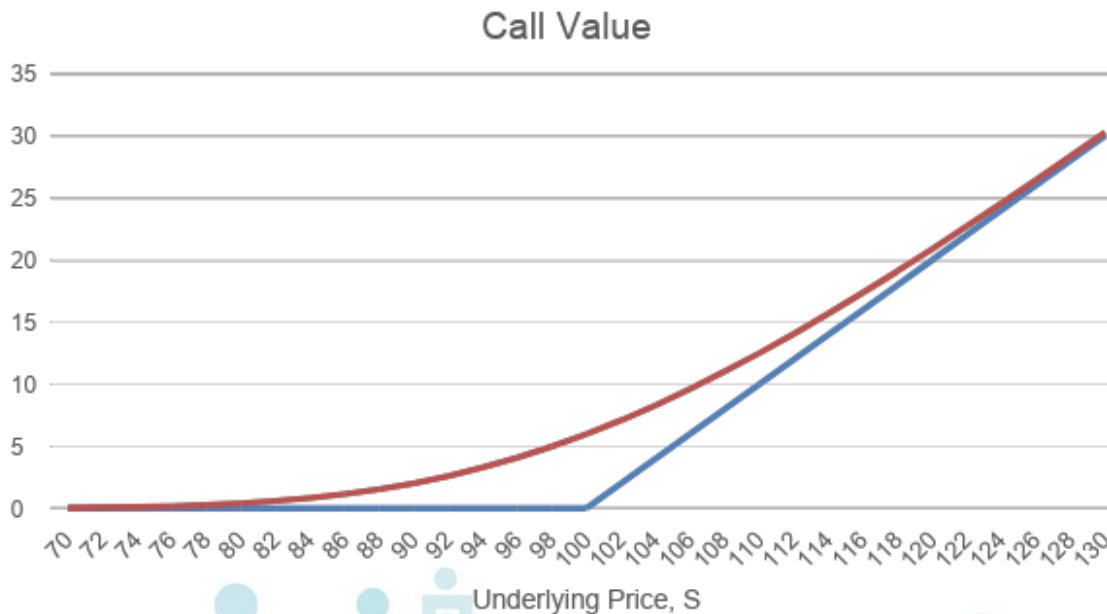
- P/C parity implies

$$\Delta_{call} - \Delta_{put} = 1$$

- So, a 90-delta call (0.9) will have the same strike as a 10-delta put (really -0.1)

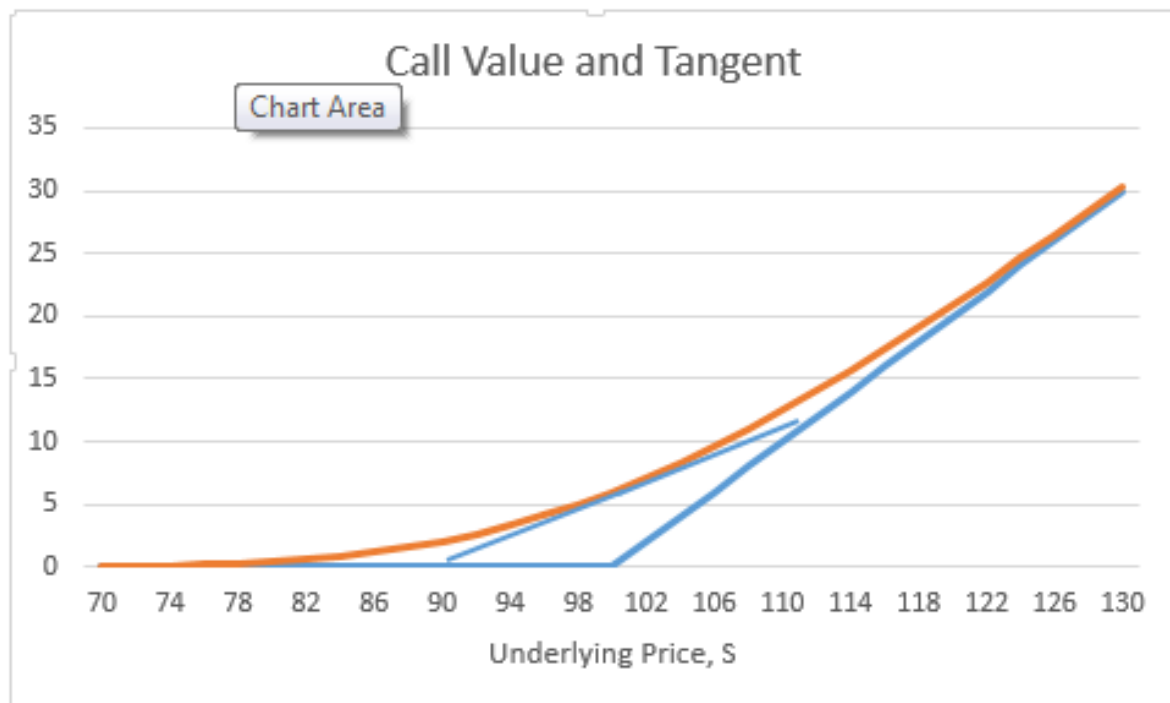
Delta (Graphically)

- Delta is the tangent of the option price Vs underlying graph.
- We have a call with a strike of 100.
- Expiration value (blue) and current premium (red) are shown below:



Delta (Graphically)

- Approximate the derivative by a difference.
- $\frac{\partial C}{\partial S} \approx \frac{\Delta C}{\Delta S}$
- This is tangent of the Call value.

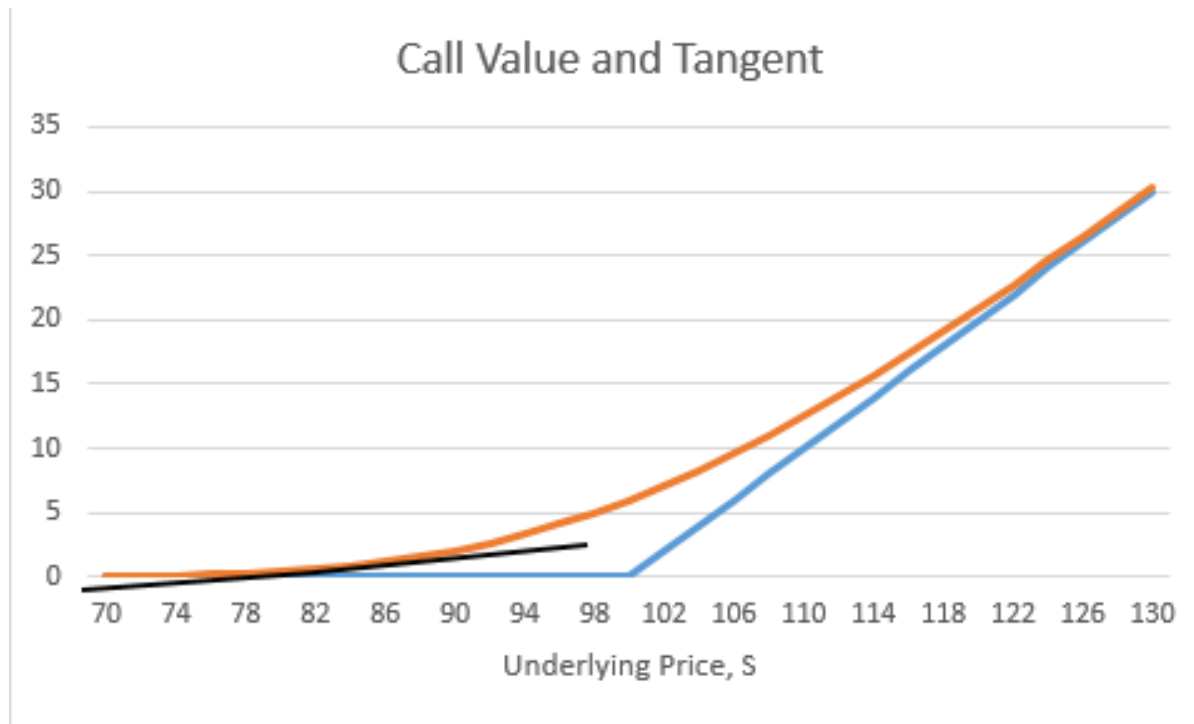


Options

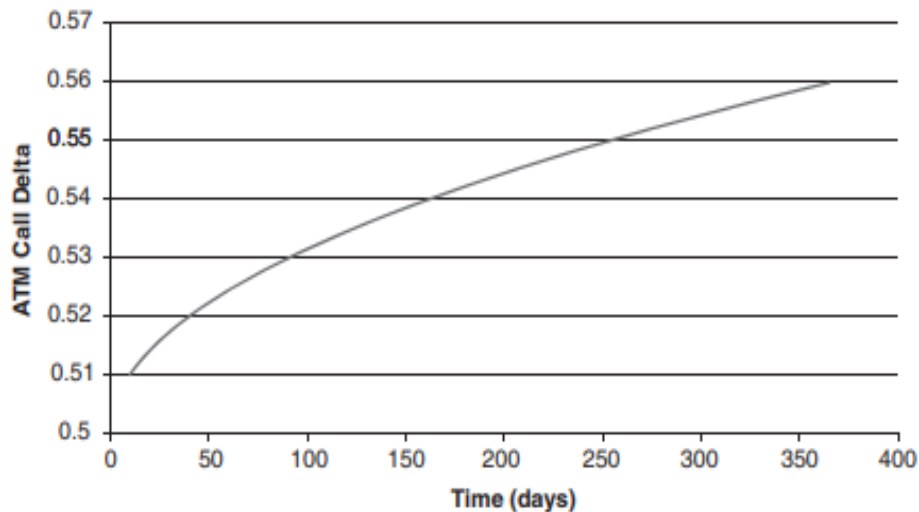
Greeks: Gamma

Delta (Graphically)

- If underlying price is much lower than the strike, the option is way out of the money and delta will be almost zero.
- Here is the delta line when $S=80$.



Delta vs Time/Volatility



- At infinite time (or volatility) call delta goes to one and put delta to zero.
- At high volatility, the stock can go much higher but is bounded below.

Delta Misconceptions

- Delta IS the partial derivative of the option price w.r.t. the underlying.
- Delta IS the hedge ratio.
- Delta IS NOT probability of finishing in the money. First, in risk neutral world this is $N(d_2)$. Second, risk neutral world isn't the real world.
- ATM delta is close to 0.5 but calls will be slightly higher.

Delta Misconceptions

- ATM call delta >0.5 and increases as volatility increases.
- Why?
- Normal returns mean percentage moves are of equal probability up or down.
- $S=100$, two 1% up moves $100 \rightarrow 101 \rightarrow 102.1$ i.e., up \$2.1
- $S=100$, two 1% down moves $100 \rightarrow 99 \rightarrow 98.1$ i.e., down \$1.9
- So as delta is the hedge-ratio we need to sell more shares to protect against a down move because it will be a smaller move.
- If volatility is much higher the effect is magnified, so delta of a call increases with volatility.

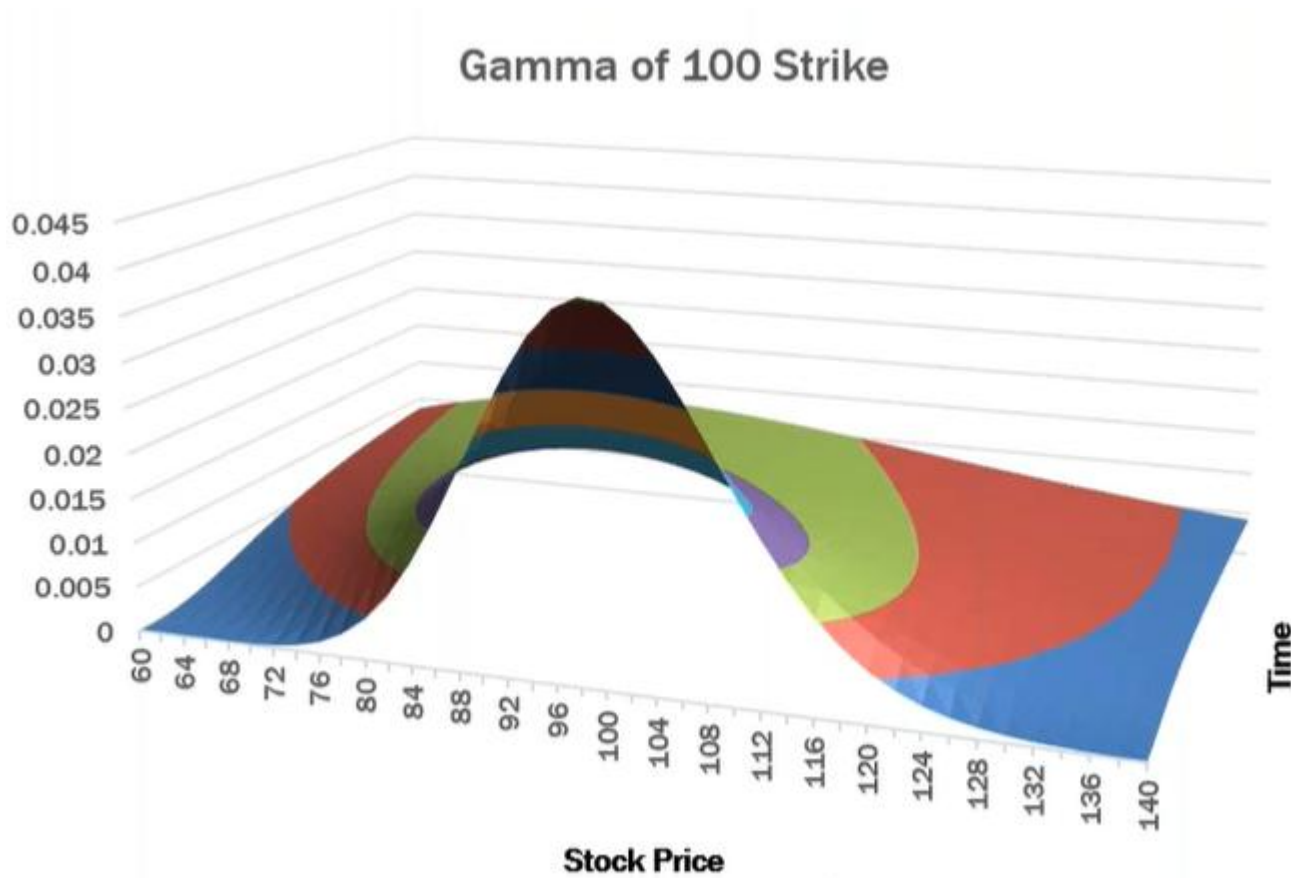
Gamma

- Gamma is the partial derivative of the delta w.r.t. the underlying.
- To manage delta, must know gamma.

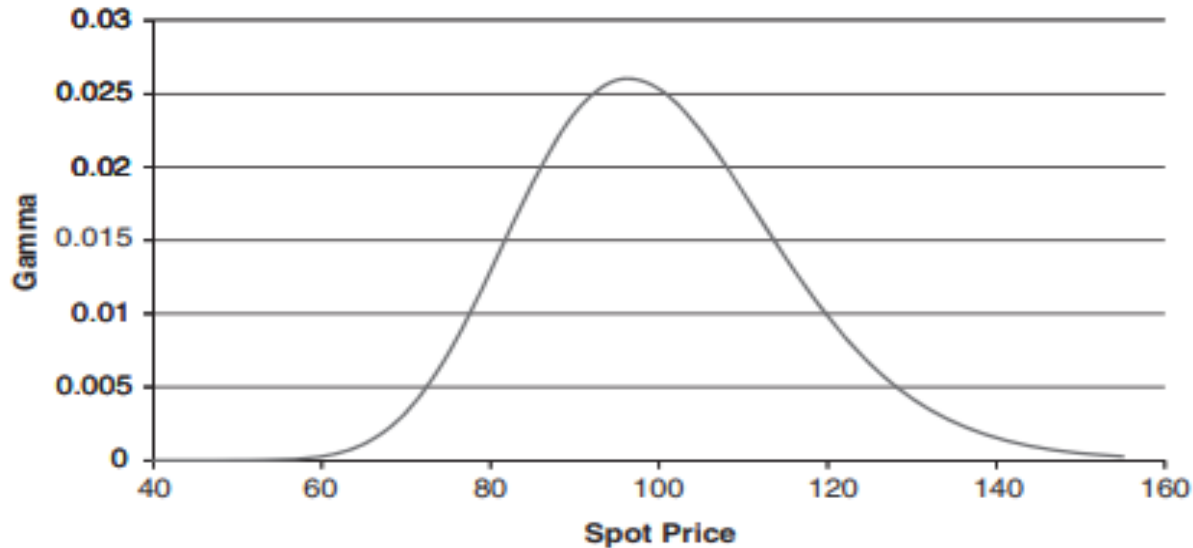
$$\Gamma_{call} = \frac{n(d_1)}{S\sigma\sqrt{t}} = \Gamma_{put}$$

- Gamma is highest for ATM short-dated options (actually slightly below the actual ATM strike).
- Indeed, traders usually refer to short dated options just as “gamma”.

Gamma

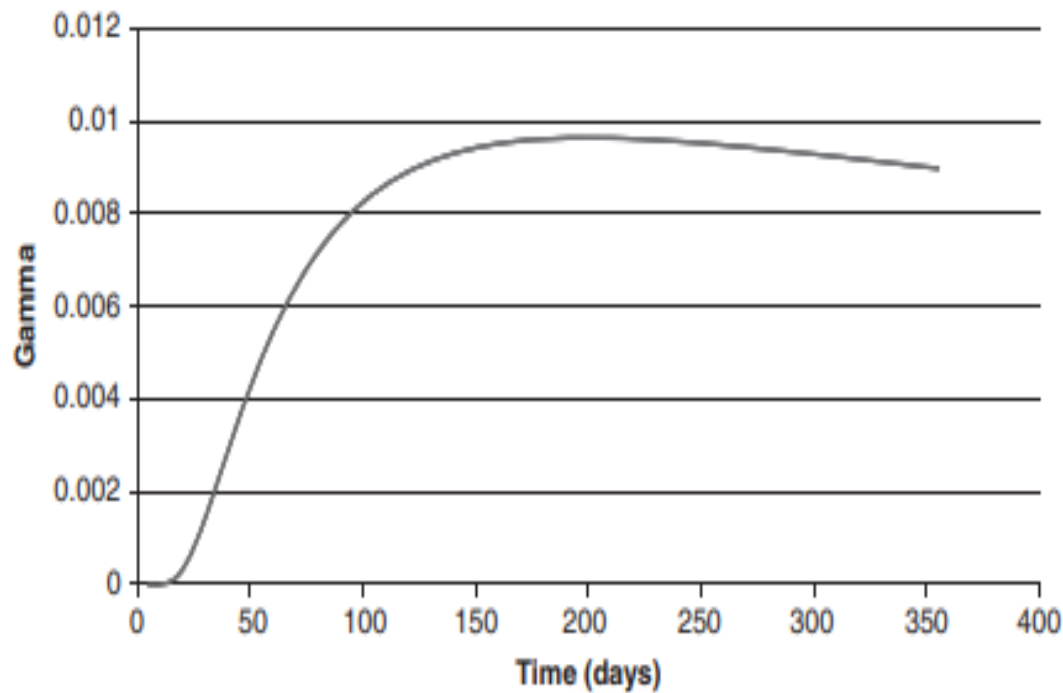


Gamma V Stock Price

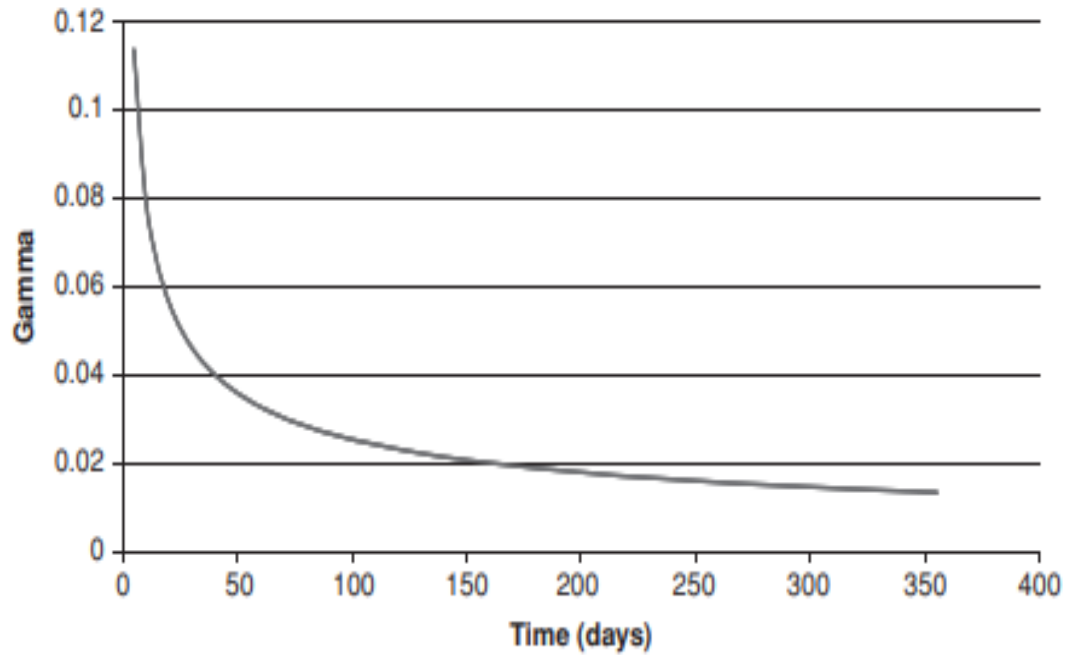


- Unsurprisingly, delta looks a lot like a cumulative normal distribution and gamma looks a lot like a normal distribution.

Gamma V Time/Vol Out of the Money Option



Gamma V Time/Vol At the Money Option



Options

Greeks: Theta

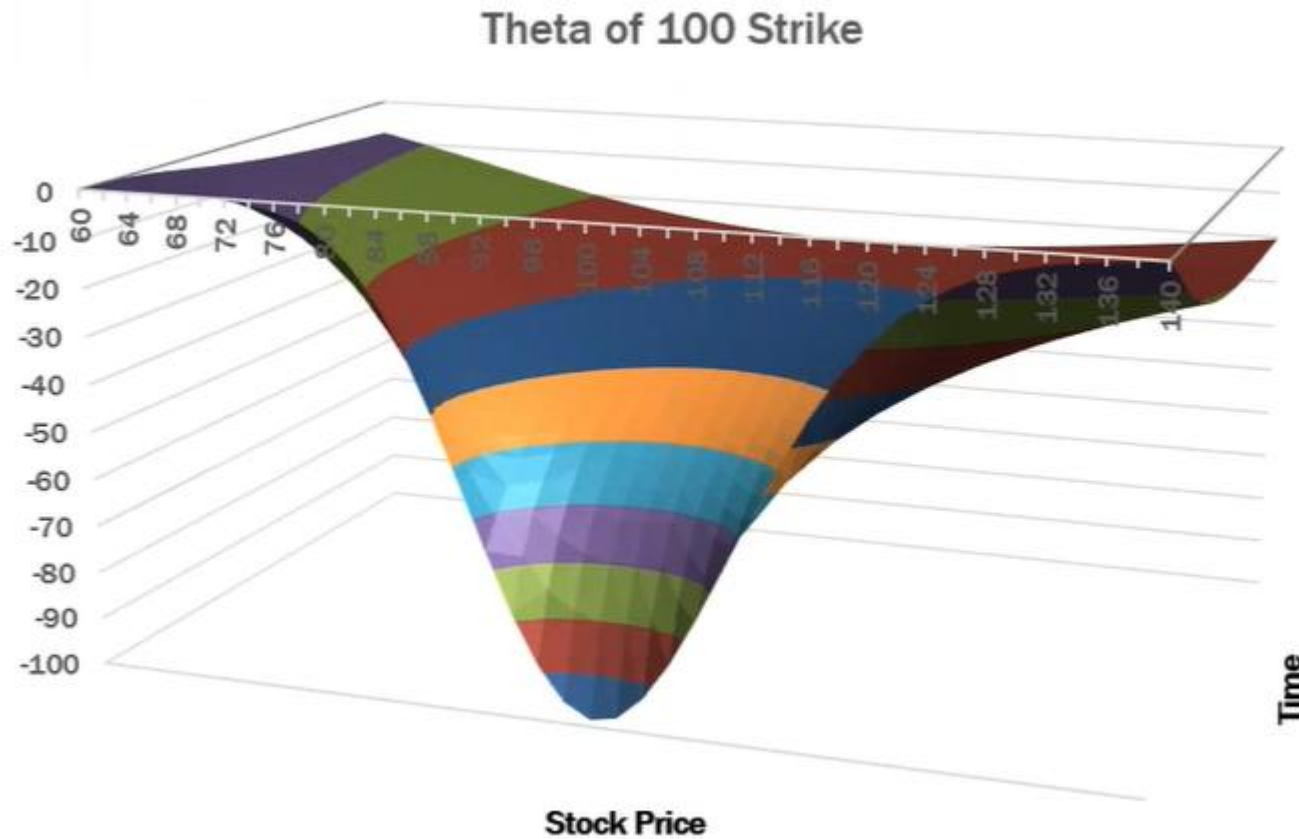
Theta

- Theta is the partial derivative of the option price w.r.t. time.
- Normally expressed in daily units.

$$\theta_{call} = -\frac{S\sigma n(d_1)}{2\sqrt{t}} - rX\exp(-rT)N(d_2)$$

$$\theta_{put} = -\frac{S\sigma n(d_1)}{2\sqrt{t}} + rX\exp(-rT)N(-d_2)$$

Theta



Theta

- First term (same for puts and calls) is the time decay, the amount the option declines because volatility has less time to act.
- Second term is often misunderstood (or just ignored).

$$\theta_{call} - \theta_{put} = rXexp(-rT)$$

- From P/C parity discussion we know a long call and short put has no volatility exposure.
- But it does have carry, the cost of holding the strike value in cash.
- Theta just allocates this carry across calls and puts according to the chance of each finishing in the money.

Options

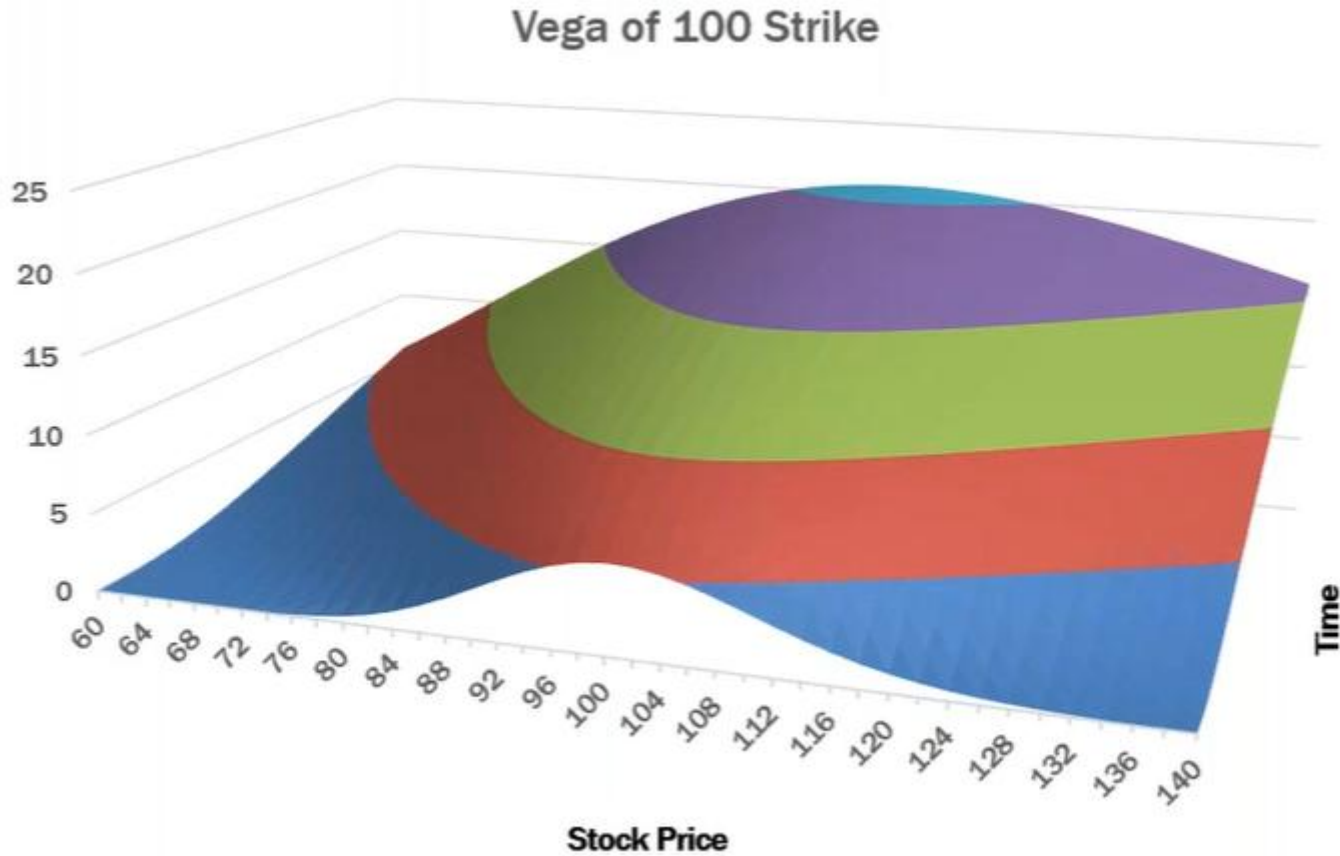
Greeks: Vega

Vega

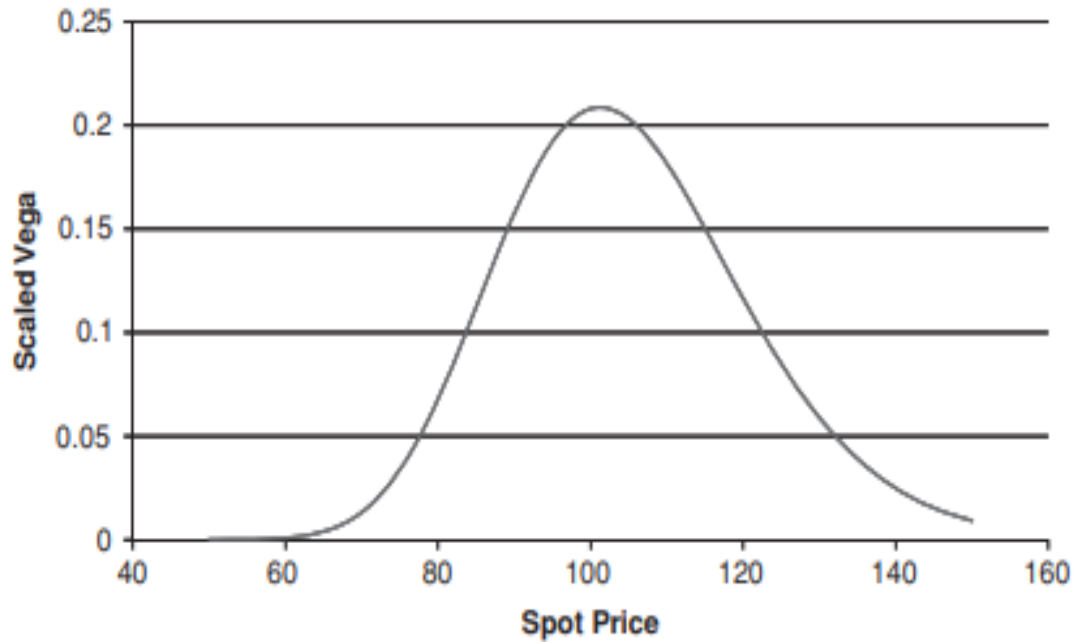
- Vega is the partial derivative of the option price w.r.t. volatility.
- Inconsistent with the assumption of constant volatility, but very important to traders.

$$Vega_{call} = Sn(d_1)\sqrt{t} = Vega_{put}$$

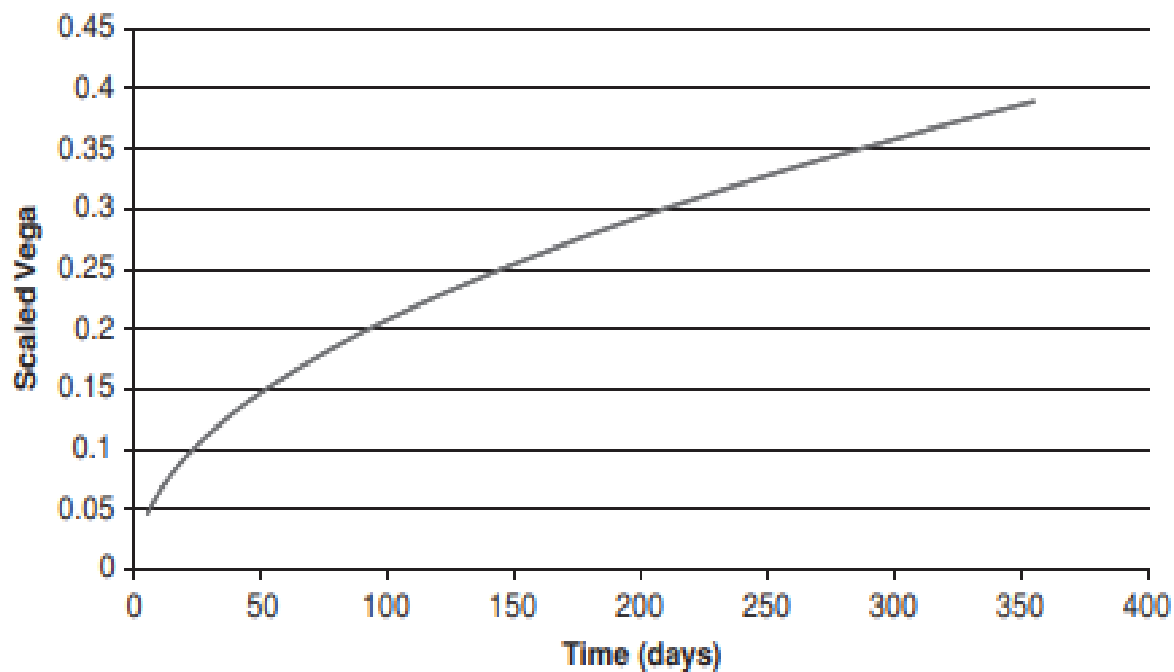
Vega



Vega Vs Spot



Vega Vs Time



Options

Greeks: Rho

Minor Greeks: Rho

- Rho is the partial derivative of the option price w.r.t. interest rates.
- Inconsistent with the assumption of constant rates.
- Usually ignored by traders, as it is managed at the firm level by treasury.
- Generally scaled to be dollar change for a 1% move in rates.

$$\rho_c = TX \exp(-rT) N(d_2)$$

$$\rho_p = -TX \exp(-rT) N(-d_2)$$

Options

Secondary Greeks

Secondary Greeks

- From just looking at equations or graphs it is clear that the greeks also have derivatives.
- E.g. the derivative of delta w.r.t. volatility, the derivative of Vega with respect to time etc.
- Many books and authors want you to think it is vital to know the exact form of all of these.
- Not really true.
- You need to know they *exist*.
- But no trading or risk management ideas are dependent on these actual equations.

Secondary Greeks

- Any professional level trading or risk system will show these greeks, but detailed understanding of their characteristics isn't really essential. (We will return to this when we look at risk management).
- B.T.W. There is no standard nomenclature for these derivatives.
- E.G. The derivative of delta with respect to volatility is variously called "vanna", "DdeltaDvol", "DdelV" or even "alpha", depending on who the trader learned from.

Greek Example

- Imagine these are my aggregate greeks:
- Long 5000 delta: I make \$5000 if the stock goes up \$1.
- Long 100 gamma: my delta changes to 5100 if the stock goes up a dollar.
- Short 500 theta: I expect to lose \$500 a day due to lose of future optionality and carry costs.
- Long 1000 vega: if volatility goes up one point, I make \$1000.

Trading Screen

DEC 31 '20	JAN 08 '21	JAN 15 '21	JAN 22 '21	MORE ▾				
14 DAYS	22 DAYS	29 DAYS	36 DAYS					
					TABBED VIEW ▾	All STRIKES ▾	SMART ▾	UVXY ▾ 100
CALLS				STRIKE	PUTS			
VEGA	IMPLIED VOL....	BID x ASK	DELTA		VEGA	IMPLIED VOL....	BID x ASK	DELTA
0.003	93.2%	3.65 x 3.70	0.959	7	0.003	93.1%	0.05 x 0.06	-0.041
0.006	100.6%	2.82 x 2.86	0.877	8	0.006	99.9%	0.21 x 0.22	-0.123
0.010	108%	2.17 x 2.20	0.757	9	0.010	110.1%	0.57 x 0.58	-0.243
0.011	122.2%	1.72 x 1.78	0.637	10	0.011	120.8%	1.11 x 1.12	-0.363
0.012	130.3%	1.41 x 1.42	0.536	11	0.012	131.8%	1.80 x 1.81	-0.464
0.012	140.2%	1.19 x 1.20	0.457	12	0.012	140.5%	2.56 x 2.59	-0.543
0.011	150.1%	1.03 x 1.04	0.395	13	0.011	151%	3.40 x 3.45	-0.605
0.010	159.6%	0.90 x 0.91	0.346	14	0.010	159%	4.25 x 4.30	-0.654
0.011	167.4%	0.81 x 0.82	0.308	15	0.011	166.1%	5.15 x 5.20	-0.692
0.009	175%	0.73 x 0.74	0.277	16	0.009	173.5%	6.05 x 6.15	-0.723
0.010	180%	0.66 x 0.68	0.250	17	0.010	182%	7.00 x 7.10	-0.750

IV: 127.2%

ON

Strategy Builder

Strategies ▾

ACTN	RT	LST TRD DAY	STRIKE	TYPE	DELTA	THETA	BID/ASK	SIZE
------	----	-------------	--------	------	-------	-------	---------	------

Order Entry

Default ▾

LMT

STP

DAY

advanced +

Account:

Submit Order